Quaternions and Octonions, from the Coordinate Plane

Start where you are: the coordinate plane

You know the coordinate plane. A point is a pair of numbers (x, y). You go x steps right and y steps up. That's it. The plane is just $\mathbb{R}^2$: pairs of real numbers.

You can already *add* points like vectors:

$$(3, 2) + (1, 4) = (4, 6).$$

Component-wise. No surprises.

But here's the question that sets up everything else:

> Can we also *multiply* two points and get a third point, in a way that's actually useful?

For real numbers on a line, multiplication is obvious: $3 \times 2 = 6$. What does $(3, 2) \times (1, 4)$ even mean? You could try multiplying component-wise — $(3 \cdot 1, 2 \cdot 4) = (3, 8)$ — but that turns out to be boring and not useful for anything. We want a multiplication rule that means something.

The 2D answer: complex numbers

Here is the cleverest possible multiplication on the plane. Pick a special point and call it i. We'll put i at $(0, 1)$ — one step up the vertical axis.

Define one rule:

$$i \cdot i \;=\; -1.$$

That single rule, plus "multiplication should behave like normal multiplication otherwise" (distributive, associative, commutative), forces *every other product*. A general point (a, b) on the plane gets written as $a + b\,i$, and then you just multiply out:

$$(a + b\,i)(c + d\,i) \;=\; ac + ad\,i + bc\,i + bd\,i^2 \;=\; (ac - bd) + (ad + bc)\,i.$$

So in coordinates:

$$(a, b) \cdot (c, d) \;=\; (ac - bd,\; ad + bc).$$

That's it. That's all complex numbers are: *the coordinate plane with that one multiplication rule.* The letter "i" is just the name of the point $(0, 1)$. There is nothing imaginary about it.

Why this rule is magic

Try multiplying any point by i:

$$i \cdot (a, b) \;=\; (0, 1) \cdot (a, b) \;=\; (-b,\; a).$$

Look at what happened: (a, b) became $(-b, a)$. That is exactly a 90° counter-clockwise rotation about the origin.

Multiplying by i *rotates the plane by* 90°.

More generally, multiplying by any complex number scales the plane and rotates it by a fixed angle. Complex multiplication *is* the language of 2D rotation-and-scaling. This is why complex numbers show up in electrical engineering, signal processing, quantum mechanics — anywhere you need to rotate things.

The summary so far:

- Real numbers $\mathbb{R}$: a 1D line.
- Complex numbers $\mathbb{C}$: the 2D plane, with a multiplication that captures rotation.

That extra structure — being able to multiply — is what makes $\mathbb{C}$ way more useful than just "pairs of numbers."

Hamilton's question (the obvious next move)

If $\mathbb{C}$ is the plane with a useful multiplication, can we do the same thing in 3D? Can we put a useful multiplication on $\mathbb{R}^3$ — triples (x, y, z) — that captures rotations in 3D space?

This question consumed William Rowan Hamilton, an Irish mathematician, for over a decade. He kept trying. He kept failing. His sons used to ask him at breakfast, "Well, Papa, can you multiply triples?" and he would say, sadly, "No, I can only add and subtract them."

The reason he was stuck: *it is impossible.* There is no useful multiplication on $\mathbb{R}^3$. You can prove it.

The breakthrough: use 4 dimensions, not 3

On October 16, 1843, Hamilton was walking with his wife along the Royal Canal in Dublin, on his way to a meeting at the Royal Irish Academy. He realised — in a flash, the way these things happen — that he didn't need three dimensions. He needed *four*.

He took his pocket-knife and carved the formula into the stone of Brougham Bridge:

$$i^2 \;=\; j^2 \;=\; k^2 \;=\; i\,j\,k \;=\; -1.$$

The carving is gone, but there is a plaque on the bridge today commemorating the moment. This was the birth of **quaternions**.

What quaternions are

A quaternion is a point in 4D space:

$$q \;=\; a + b\,i + c\,j + d\,k,$$

where a, b, c, d are real numbers and i, j, k are three special "unit" directions, each perpendicular to the others and to the "real" direction.

The set of all quaternions is called $\mathbb{H}$, after Hamilton.

The multiplication rule is everything in

$$i^2 = j^2 = k^2 = -1, \qquad ij = k, \quad jk = i, \quad ki = j.$$

(Same cyclic pattern as the right-hand rule.) And here is the twist:

$$ji = -k, \quad kj = -i, \quad ik = -j.$$

Order matters. $ij \neq ji$. Quaternions are *non-commutative*. This is the first new thing.

The bargain: you trade commutativity for an extra dimension

- Real numbers (1D): you can multiply, and $ab = ba$.
- Complex numbers (2D): you can multiply, and $ab = ba$.
- Quaternions (4D): you can multiply, but $ab \neq ba$ in general.

Every time you double the dimension, you get a useful multiplication rule, but you pay for it by giving up a property you took for granted.

What quaternions are good for

Just as multiplying by a complex number rotates the plane, multiplying by a quaternion rotates 3D space. Specifically: sandwich a 3D vector v between a quaternion q and its "conjugate" q^{-1}:

$$v \;\mapsto\; q\,v\,q^{-1}.$$

The result is v rotated. The quaternion q encodes the axis and angle of rotation.

This is not a toy. Quaternions are how:

- Video games and 3D animation software smoothly rotate objects without "gimbal lock" (the failure mode of just using three angles).
- Aerospace and robotics track orientation.
- Physicists describe the spin of electrons (via the closely related *Pauli matrices*).

If you have ever rotated a camera in Unity or Unreal or Blender, quaternions did the work.

Doing the trick again: octonions

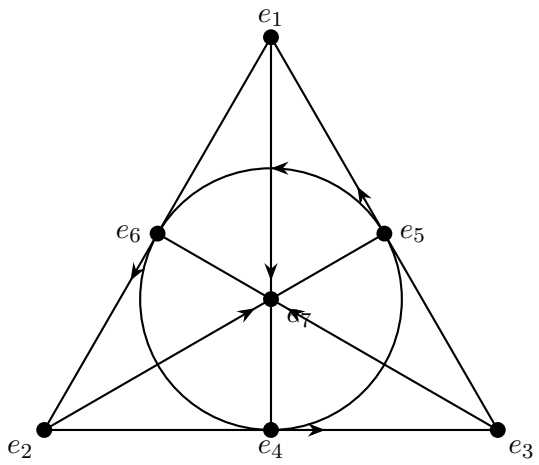
We doubled the dimension from 1 to 2 (real $\to$ complex) and got rotations of the plane. We doubled from 2 to 4 (complex $\to$ quaternion) and got rotations of 3D space. There is a recipe — the *Cayley–Dickson construction* — for doing this doubling mechanically. So: what happens if we double again?

You get an 8-dimensional algebra called the **octonions**, written $\mathbb{O}$. Discovered by John Graves in late 1843 (just two months after Hamilton's quaternions) and independently by Arthur Cayley in 1845.

An octonion is a point in 8D space, with one real direction and seven "imaginary" directions $e_1, e_2, \ldots, e_7$. They each square to -1, they each anticommute with the others, and the products $e_i e_j$ follow a specific rule.

But what is the rule? Writing out all $7 \times 7 = 49$ products in a table is awful. There is a much better way to present them: the *Fano plane*.

The Fano plane: octonion multiplication in one picture



What you are looking at. The Fano plane is the smallest projective plane. It has exactly:

- 7 points (the 7 imaginary units $e_1, \ldots, e_7$);
- 7 lines, each containing exactly 3 points;
- the property that any two points lie on a unique line.

Six of the lines are drawn straight; the seventh is the circle through the three edge-midpoints. "Line" here means "line of the projective plane" — the circle is geometrically a circle but combinatorially a line.

How to read it as a multiplication table. Each line carries an orientation (the arrows). If a line points $e_a \to e_b \to e_c \to e_a$ cyclically, then

$$e_a\, e_b \;=\; e_c, \qquad e_b\, e_c \;=\; e_a, \qquad e_c\, e_a \;=\; e_b,$$

and going *against* the arrow flips the sign:

$$e_b\, e_a \;=\; -e_c, \quad e_c\, e_b \;=\; -e_a, \quad e_a\, e_c \;=\; -e_b.$$

Together with $e_i^2 = -1$, that is the entire octonion multiplication.

The seven lines spelled out. For our labelling above:

Line	Oriented triple	Products
bottom side	$e_2 \to e_4 \to e_3 \to e_2$	$e_2e_4 = e_3,\ e_4e_3 = e_2,\ e_3e_2 = e_4$
right side	$e_3 \to e_5 \to e_1 \to e_3$	$e_3e_5 = e_1,\ e_5e_1 = e_3,\ e_1e_3 = e_5$
left side	$e_1 \to e_6 \to e_2 \to e_1$	$e_1e_6 = e_2,\ e_6e_2 = e_1,\ e_2e_1 = e_6$
median from e_1	$e_1 \to e_7 \to e_4 \to e_1$	$e_1e_7 = e_4,\ e_7e_4 = e_1,\ e_4e_1 = e_7$
median from e_2	$e_2 \to e_7 \to e_5 \to e_2$	$e_2e_7 = e_5,\ e_7e_5 = e_2,\ e_5e_2 = e_7$
median from e_3	$e_3 \to e_7 \to e_6 \to e_3$	$e_3e_7 = e_6,\ e_7e_6 = e_3,\ e_6e_3 = e_7$
inscribed circle	$e_4 \to e_5 \to e_6 \to e_4$	$e_4e_5 = e_6,\ e_5e_6 = e_4,\ e_6e_4 = e_5$

Worked example. To compute $e_3 \cdot e_5$: find the unique line through e_3 and e_5 — the right side of the triangle. The orientation is $e_3 \to e_5 \to e_1$, and the arrow goes from e_3 *to* e_5, so

$$e_3e_5 \;=\; e_1.$$

To compute $e_5 \cdot e_3$ instead: same line, against the arrow, so $e_5e_3 = -e_1$.

Why the Fano plane?

There is something marvellous in the fact that octonion multiplication is encoded in a tiny combinatorial object. The Fano plane is one of the most studied objects in finite geometry: it is the projective plane $\mathrm{PG}(2, \mathbb{F}_2)$ over the field with two elements. The same picture shows up in the theory of error-correcting codes (the binary $[7, 4]$ Hamming code), in the structure of the simple group $\mathrm{PSL}_2(\mathbb{F}_7)$, and now in the multiplication table of the octonions. It is the smallest projective plane that exists at all — and it carries exactly enough structure to define an 8-dimensional division algebra. That coincidence is part of why the octonions feel miraculous.

The next bargain

- Real numbers $\mathbb{R}$ (1D): commutative, associative.
- Complex numbers $\mathbb{C}$ (2D): commutative, associative.
- Quaternions $\mathbb{H}$ (4D): *not commutative*, but still associative.
- Octonions $\mathbb{O}$ (8D): not commutative *and not associative*.

Associative means $(a \cdot b) \cdot c$ equals $a \cdot (b \cdot c)$ — you can group the parentheses however you want. With octonions, this fails: $(ab)c$ and $a(bc)$ can be different octonions. Almost every algebraic tool you have ever used quietly assumed associativity. Without it, normal calculation becomes treacherous.

Why stop at 8?

Try doubling again. You can. The result is the 16-dimensional "sedenions." But now the bargain has changed: the sedenions have *zero divisors*. You can find two nonzero sedenions whose product is zero. That means you can't divide, and the structure breaks badly. Past dimension 8, the doubling trick stops giving you something useful.

This is a theorem, not a fashion:

Hurwitz's theorem (1898). The only *normed division algebras* over $\mathbb{R}$ have dimensions 1, 2, 4, or 8.

In plain terms: real numbers, complex numbers, quaternions and octonions are the *only* ways to put a well-behaved multiplication on $\mathbb{R}^n$. There is no 5D or 7D version. There is no 16D version. The four are the whole list.

Even more remarkable: a much harder result (Frobenius, with later work by Bott, Milnor, Kervaire and Adams) connects these dimensions to deep topological facts about spheres. The numbers $1, 2, 4, 8$ are not a coincidence; they're the same dimensions in which the n-sphere can be "parallelised" (combed without a cowlick). The fact that there are exactly four normed division algebras is secretly the same fact as the fact that you can comb hair flat on S^1, S^3 and S^7 but not on any other sphere.

The cheat sheet

Algebra	Dim.	What you give up
$\mathbb{R}$ (real numbers)	1	nothing
$\mathbb{C}$ (complex numbers)	2	ordering (no "which is bigger")
$\mathbb{H}$ (quaternions)	4	commutativity ($ab \neq ba$)
$\mathbb{O}$ (octonions)	8	associativity ($(ab)c \neq a(bc)$)

Why octonions are still interesting if you can't even multiply with parentheses

Octonions are weird, and for a long time they were considered a curiosity. But over the last fifty years they have shown up in several deep places:

- The five *exceptional Lie groups* (G_2, F_4, E_6, E_7, E_8) — bizarre symmetry groups that have no infinite-family explanation — are all built from the octonions. G_2 is literally the automorphism group of $\mathbb{O}$.
- In particle physics, octonions show up in proposed descriptions of fundamental particles (the "three generations" of fermions match the $\mathbb{O}$ structure suspiciously well), and in string theory and M-theory, the special role of dimensions 10 and 11 is closely tied to the octonions.
- In number theory and lattice theory, the densest known sphere packings in 8 and 24 dimensions (E_8 and the Leech lattice) are built using octonionic arithmetic.

So the joke is on us: the algebra that seemed least useful for ordinary calculation turns out to underlie some of the most exceptional and beautiful structures in mathematics.

Practical aside: can you use this in Vulkan?

Vulkan is a graphics API. The question is fair: if quaternions rotate 3D space and octonions are the next step up, can a shader or a graphics engine actually *use* octonions for anything?

Quaternions: yes, everywhere

Quaternions are the standard way to represent and compose rotations in real-time graphics, and Vulkan is no exception. In practice:

- You store a unit quaternion as a `vec4` on the CPU side and pass it to a shader as a uniform or vertex attribute. GLSL/SPIR-V has no native quaternion type, but you only need a handful of helper functions: quaternion multiplication, conjugate, and the rotation sandwich $v \mapsto q\,v\,q^{-1}$ (which simplifies for unit q).
- Skeletal animation systems store bone orientations as quaternions because they interpolate well: spherical linear interpolation ("slerp") gives smooth, constant-speed rotation between two orientations, with no gimbal lock and no normalisation drift problems that Euler angles would have.
- Cameras, IMU integration, and orientation tracking all use quaternion arithmetic for the same reasons.

If you write a Vulkan renderer, you will end up writing or importing a quaternion library. That is essentially mandatory.

Octonions: effectively no, and there is a real reason

Octonions cannot replace quaternions for 3D rotation, and there is nowhere else in a standard graphics pipeline that they slot in. The specific obstacles:

- **Non-associativity wrecks composition.** The reason quaternions are useful is that composing two rotations is the same as multiplying their quaternions: rotate-by-q_1 then rotate-by-q_2 becomes rotate-by-q_2q_1. Octonion multiplication is *not* associative, so $(q_1q_2)v(q_1q_2)^{-1}$ is not the same as $q_1\big(q_2vq_2^{-1}\big)q_1^{-1}$. You cannot "stack" octonion rotations the way you do quaternion rotations, because the algebra refuses to cooperate.
- **Unit octonions are not a group.** Unit quaternions form the sphere S^3, which is a Lie group, which is exactly what rotations need. Unit octonions form S^7, which is famously *not* a group — it is a Moufang loop, a weaker structure. There is no group of 3D rotations sitting inside the octonions.
- **3D space doesn't fit.** The natural geometric home of octonions is in dimensions 7 and 8 (where they actually do encode special rotations, via the exceptional groups G_2, Spin(7) and Spin(8)). Graphics pipelines work in dimension 3, where the octonionic structure is, simply, the wrong tool.

So where might octonions plausibly appear?

Honest answer: nowhere in the mainline rendering pipeline. The places they do appear are research-flavoured and far from a typical Vulkan workload:

- *Octonionic and Clifford neural networks.* Some research groups have built networks whose weights live in $\mathbb{H}$ or $\mathbb{O}$, motivated by signal-processing problems with natural multi-channel structure. These can in principle run on a GPU, but not via Vulkan's graphics path — they look like ordinary tensor workloads to a compute backend.

- *Geometric algebra / Clifford algebra renderers.* A small community uses geometric algebra ($\mathrm{Cl}_{3,0}$, $\mathrm{Cl}_{3,1}$, etc.) as a unifying framework for graphics math. These algebras are cousins of the quaternions and octonions, but the actual workhorses are the Clifford algebras, not $\mathbb{O}$ itself.
- *Exotic research demos.* You can write a shader that evaluates an octonion fractal (the analogue of the Mandelbrot set in $\mathbb{O}$), or that visualises subsets of 7-dimensional space using octonionic structure. These are beautiful and useless, in the best mathematical tradition.

Bottom line. In Vulkan, quaternions are essential; octonions have no native role in 3D rendering. The reason is structural — non-associativity — not a missing library.

OK fine — here is how to actually put octonions in Vulkan

You can absolutely put octonions in a Vulkan shader. The issue is not "can you", it is "what do you do with them." Octonions are 8 floats and a multiplication rule; GPUs eat 8 floats for breakfast. Here are the genuinely viable things to do:

1. Octonion fractals (the fun answer). Just as the Mandelbrot set lives in $\mathbb{C}$ and quaternion Julia sets live in $\mathbb{H}$, you can iterate $z \mapsto z^2 + c$ in $\mathbb{O}$ and get *octonion fractals*. Since the result lives in 8D, you take a 3D slice (fix five of the eight coordinates) and render it as a volumetric or signed-distance object in a fragment shader. This is the closest thing to a "real" Vulkan use of octonions, and people do render these.

2. Octonion neural networks (compute shader workloads). Some research uses octonion-valued weights for signal-processing tasks with natural 7- or 8-channel structure (multi-spectral imagery, certain audio problems). Vulkan compute shaders are a perfectly reasonable backend; you're not using the graphics pipeline, but you are using Vulkan.

3. Procedural noise / high-dimensional sampling. The seven imaginary units give you a 7D direction space with rich algebraic structure. You can use it as the substrate for a procedural noise function with built-in symmetries you would not get from $\mathbb{R}^7$.

The GLSL skeleton. Define an octonion as a `vec4 + vec4` pair (real part plus seven imaginary components, packed naturally on 16-byte boundaries):

```
struct Oct {
    float r;       // real part
    float e[7];    // e_1 ... e_7
};

// Multiplication using the Fano-plane rules.
// Each pair (e_i, e_j) is on exactly one line; the line tells you
// which e_k you get and with what sign. The 21 sign rules below
// come directly from the seven oriented triples in the Fano plane.
Oct omul(Oct a, Oct b) {
    Oct c;
    c.r = a.r*b.r - a.e[0]*b.e[0] - a.e[1]*b.e[1] - a.e[2]*b.e[2]
                  - a.e[3]*b.e[3] - a.e[4]*b.e[4] - a.e[5]*b.e[5]
                  - a.e[6]*b.e[6];
```

```
    // imaginary parts: scalar*imag + imag*scalar + cross-products
    // following the Fano lines. (Expand the 7 lines into 7*3 = 21
    // pairwise products, with anticommutativity giving the rest.)
    // ... (~50 lines of straight-line arithmetic)
    return c;
}
```

The expansion of the cross-products is mechanical from the table of seven Fano lines given earlier. There is no branching, no recursion, no loops — just a fixed pile of multiply-adds, which is exactly what a GPU executes fastest.

Octonion fractal fragment shader, in outline.

```
// for each pixel:
Oct c = octFromUV(uv);              // pick c from screen + sliders
Oct z = octZero();
for (int i = 0; i < MAX_ITER; ++i) {
    z = oadd(omul(z, z), c);
    if (onorm2(z) > 4.0) break;
}
// shade based on i (escape time) or use a distance estimator
```

What this won't do. It will not make your rotations better, replace your quaternion library, or give you any new graphics primitive. What it *will* do is let you render objects nobody else has, because the slice of $\mathbb{O}$ you choose to project is yours to pick. If "octonions in Vulkan" is the project, the octonion fractal viewer is the actual thing to build.

Recap, in one breath

The coordinate plane is just $\mathbb{R}^2$ until you give it a multiplication rule; then it becomes $\mathbb{C}$, and that rule is exactly 2D rotation. Doubling the dimension gives $\mathbb{H}$, which encodes 3D rotation, at the cost of $ab \neq ba$. Doubling once more gives $\mathbb{O}$, which gives up associativity too. Doubling beyond that breaks. The four algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ are the complete list of well-behaved multiplications you can put on $\mathbb{R}^n$, and the same four dimensions $1, 2, 4, 8$ show up in topology, Lie theory and physics in ways nobody fully understands.

Further reading

- John Baez, "The Octonions," *Bulletin of the AMS* **39** (2002), 145–205. The standard friendly survey; readable by undergraduates, deep enough for researchers.
- John Stillwell, *Naive Lie Theory*, Springer, 2008. Beautiful book that builds up to quaternions and beyond from the coordinate plane.
- John H. Conway and Derek A. Smith, *On Quaternions and Octonions*, A. K. Peters, 2003. Short, authoritative, readable.
- Marcus du Sautoy, *Finding Moonshine*, Fourth Estate, 2008. Includes a chapter on quaternions and the bridge story.